BATCH6_RTA_GEOM

PREREG: BATCH6_RTA_GEOM (Map-C21 to Map-C24): the 56 P^+ same-tower triples and the chord-zero path-H^2 proxy over the 8 fibres

Written 2026-10-24 at about 7:10 PM CDT, before any BATCH6 test. Owner: Arvin Hampton. New files only. CORE_FREEZE. All maps are **Category B**. A pass is a survived prediction, not a bridge; a pass flagged vacuous carries no weight. order21 and the paper are not touched. C15–C20 are not reopened. No Π, grading or predicate is invented. Runs: nice, OMP_NUM_THREADS=1, ulimit -v 3000000.

Written objects (computed by importing the source probe's own functions)

Source: /tmp/repo539/docs/S2-11DM2ET-X-Axiomatic-Book/scripts/residual_p_plus_multiscale_probe.py. Functions atoms() and tower_of() are imported read-only. - O: the probe's core, O = residual[i] with i % 9 == 0 (this equals 21+9j). - Triples: all 3-subsets (i,j,k) of core indices in one tower bucket that has ≥ 3 elements (written count 56). - Window: W = 18. The written split is 3 in the window (k < 18) and 53 in the tower region (i ≥ 17). - Chord-zero matrix A (56 × 538): row t has a 1 at column i if j = i+1, and a 1 at column j if k = j+1. Proxy = 56 − rank A. - Gates (else VOID): 56 triples; in-window count 3; in-tower count 53. - Fibre of a triple: β♯ of its **start index** i, i.e. i mod 8. This is an operational convention fixed here; the sources do not assign triples to fibres. - Per-fibre proxy: proxy_r = n_r − rank(A restricted to rows with i ≡ r).

Modular data (written): P-odd/P-even symbol split 51/54 (C3_RUN post-hoc); $\chi_{140}(2r+1)$ for r = 0..7 is [1, 1, 0, 0, 1, 0, 1, 1]; the three unary-theta norms 5/2, 1/34, 35/4 (three distinct thetas).

Map-C21 (51/54 split → parity of fibres)

Prediction: let E = #triples with even start fibre and Od = # with odd start fibre (E + Od = 56). Then Od < E and E − Od ≤ 3 (mirroring 54 − 51 = 3). PASS iff both hold.

Map-C22 (χ_{140} → forbidden fibres)

Prediction: no triple starts in a fibre r with $\chi_{140}(2r+1) = 0$, i.e. $r \in \{2, 3, 5\}$. PASS iff $n_2 = n_3 = n_5 = 0$.

Map-C23 (RG1′ zero-set “iff” structure → chord-zero path H^2)

Prediction: proxy_r = 0 for every fibre r, and the global proxy = 0. **Flagged in advance as possibly vacuous:** if the 56 triples are pairwise disjoint consecutive runs, then A has full row rank by construction, and a pass is vacuous (Category A). The script records disjointness.

Map-C24 (three unary thetas → the 3 window triples)

Prediction: the 3 in-window triples have pairwise distinct start fibres i mod 8. PASS iff the three start residues are distinct.

Still OPEN (blocked): C6, C10, C11, E1–E3.

RESULTS: BATCH6_RTA_GEOM

PREREG.md sha256 0f19ed3d0168cee1c670b3e5f955b2003fde17e08333d4d9e823a3a36e2bc49c, hashed before tests. Script scripts/b6.py (imports the source probe’s atoms()/tower_of() read-only; hash in scripts/SHA256SUMS); data data/b6.json; log logs/b6.log. Exact (sympy ranks), under 1 s.

Gates: 56 triples, 3 in the window, 53 in the tower region. All OK (the written counts are reproduced). Triple start fibres (i mod 8), counts r = 0..7: **[13, 0, 1, 14, 13, 1, 1, 13]**. E = 28, Od = 28. Window triples: (0,1,2), (7,8,9), (14,15,16), with starts mod 8 = 0, 7, 6. All 56 triples are consecutive runs (i, i+1, i+2) and pairwise disjoint. Chord-zero proxy: global 0, and 0 in every fibre.

Map	Prediction	Measured	Verdict	Vacuous?
C21 (51/54)	Od < E and E − Od ≤ 3	E = Od = 28	**FAIL**	—
C22 (χ_{140})	$n_2 = n_3 = n_5 = 0$	$n_2 = 1$, $n_3 = 14$, $n_5 = 1$	**FAIL** (see datum error)	—
C23 (RG1′ → path H^2)	proxy_r = 0 for all r, global 0	all 0	PASS	**yes, vacuous** (flagged in the PREREG: disjoint consecutive rows ⇒ full

Map	Prediction	Measured	Verdict	Vacuous?
				row rank, Category A)
C24 (three thetas → window triples)	distinct start fibres	0, 7, 6	PASS	**yes, carrier-forced** (see below)
C6, C10, C11, E1–E3	—	blocked	OPEN	—

Bridge: NO. The two passes are vacuous or carrier-forced; the two discriminating maps failed.

Post-hoc analysis (verdicts unchanged)

- **C24 is carrier-forced (Category A).** The window triples start at i = 0, 7, 14: the core step 9 against tower width 21 puts 3 core points in a tower every 7 indices (63 = 3·21) in the 21-wide region. 7k mod 8 for k = 0, 1, 2 gives 0, 7, 6, which are always distinct. No modular datum enters, so the pass is not evidence.
- **C22 datum error (disclosed).** The PREREG transcribed $\chi_{140}(2r+1)$ as [1,1,0,0,1,0,1,1]. The correct values (sympy) are **[1, −1, 0, 0, 1, −1, 1, 0]**, so the true zero set is {2, 3, 7}, not {2, 3, 5}. C22 is recorded FAIL under its literal rule. The intended χ-tied map would also fail ($n_7 = 13 \neq 0$). No rescue; the hashed PREREG is not edited.
- The fibre profile [13,0,1,14,13,1,1,13] concentrates on $r \in \{0,3,4,7\}$, because triples are spaced by 7 or 8 on the core. This is carrier arithmetic again. # NOTES: BATCH6_RTA_GEOM ## Sources re-read
- /tmp/repo539/docs/S2-11DM2ET-X-Axiomatic-Book/scripts/residual_p_plus_multiscale_probe.py: lines 60–76 (stokes_chord_zero), 80–112 (core, buckets, the 56 triples, window split), 196–213 (chord-zero A, path-H^2 proxy). atoms()/tower_of() are imported unchanged.
- /tmp/repo539/docs/S2-11DM2ET-X-Axiomatic-Book/P_plus_Permanent_Class_Survival_Optional_Lock_2026-08-08.md (56 triples: 3 in W = 18, 53 in towers).
- BATCH5 NOTES (survival sources); BATCH3/BATCH4 NOTES ($x_j = 21+9j$, $\beta\sharp = j \bmod 8$).

Disclosures

- Fibre of a triple = start index mod 8 is an operational convention, fixed in the PREREG.
- The χ_{140} table in the PREREG was miscomputed by hand before hashing. Correct values are in RESULTS. Future preregs should compute data with a hashed script before writing them in.
- The path-H^2 chord-zero proxy is identically 0 (disjoint consecutive triples), so it cannot discriminate fibres. Like the BATCH5 survival test, it is too coarse.

Pattern across BATCH3–6 (for Arvin)

Every written residual invariant tested so far (β♯ counts, τ occupancy, permanent-class survival, P^+ triples, chord-zero H^2) is governed by the carrier arithmetic: step 9 vs tower widths 21/20, and $B' \equiv 3 \bmod 8$. The modular data (51/54, χ_{140}, theta norms, A2c support) have never predicted a non-vacuous feature. This is consistent with the Residual-vs-Resonant lock: no bridge.

Sharpest next step

- On the residual side, the cheap written invariants are exhausted. Remaining options:
  - a. A written invariant that is *not* a function of the carrier arithmetic alone. The candidates are the form spectral sequence pages or the K^+ (V = 539, E = 594, F = 56) cell structure. A next prereg could predict a K^+ Betti/torsion count over F_p for p | 140 versus p ∤ 140 from the modular datum.
  - b. Wait for a written 539-step Π / ±1 grading (C6/C10/C11/E1–E3).
- order21 and the paper were not touched. CORE_FREEZE; new files only; no push, no sends.